

Geonho Roh

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EDUCATION

B.S. Computer Sciences

University of Wisconsin-Madison

Current GPA: 3.82 / 4.0

SKILLS & INTERESTS

Languages & Frameworks: Node.js, Next.js, React.js, Swift, Dart/Flutter, Java, Kotlin, React Native, Python, PostgreSQL, MongoDB, FastAPI, Crawling, Node.js, React.js, C/CPP, Type/JavaScript,

Tools & Technologies: AWS EC2, AWS S3, AWS SES, Git, GitHub, Docker, Vercel, GCC/GDB, Linux, REST APIs, WebSockets, Web Bluetooth, QR Code, Web Crawling/Scraping

WORK EXPERIENCE

The Growth, Intern, Software Engineer

Aug 2025 – Feb 2026

- Led the development of an SCM system using **Python (FastAPI), Redis, MongoDB, and Google Sheets API**, automating **80% of manual operational workflows** and significantly improving supply chain data efficiency.
- Developed an automated marketing data collection and analysis pipeline using **Python and Selenium**, scaling data extraction volume by **over 100x** compared to manual processes.
- Led the architecture and development of an e-commerce management platform using **Node.js, MongoDB, and REST APIs**, seamlessly integrating multi-channel operations across Amazon, Shopify, and 3PL platforms.
- Spearheaded the development of a marketing email dispatch system using **Node.js, MongoDB, and Amazon SES**, optimizing campaign delivery performance and system reliability.
- Managed internal cloud infrastructure and built automated CI/CD pipelines using **Jenkins and Docker**, streamlining code deployment and maintaining service stability across multiple projects.
- Engineered custom automation utilities and scripts to streamline routine technical tasks, boosting cross-functional operational efficiency.
- Participated as a technical interviewer to evaluate candidate skills and culture fit, contributing to the hiring and onboarding of high-performing engineering talent.

Zipper, Co-Founder, Head of Software Engineering

Oct 2023 – Jan 2025

- Co-founded the company and led core end-to-end technical development, successfully raising **\$75,000 in seed funding**.
- Represented the company as a technical lead at **CES 2025 in Las Vegas**, showcasing product technology to global industry partners and investors.
- Developed and published two iOS applications—"Zipper: Collage Your Life" (social media/design) and "Zipper: Create your Card"—on the App Store using **SwiftUI, UIKit, MVVM, and REST APIs**.
- Built a high-performance Image Segmentation server using **FastAPI and SAM2 (Segment Anything Model 2)** to automate automated visual asset processing.
- Engineered a dedicated Apple Push Notification Service (APNs) server using **Next.js, PostgreSQL, and REST APIs** to power real-time user engagement.
- Constructed an automated web scraping pipeline using **Python (Selenium) and AWS S3**, extracting and processing **10,000+ fashion product datasets**.

ROK ARMY, Sergeant, Radar Operator

July 2022 – Jan 2024

- Served as a **Squad Leader (Sergeant)**, leading a tactical team while operating high-precision counter-battery radar systems (**Arthur-K**)

PROJECTS & CLUB

Personal Website | Next.js, PostgreSQL

- Architected and deployed a personal website using **Next.js and PostgreSQL**, hosting via **Vercel** with automated continuous integration (CI/CD) pipelines.
- Peer-to-peer file transfer using **WebRTC**, PDF and QR utility tools, BGP propagation map, REST APIs for game updates etc

DOT_NET (Club President) | PHP, Dart/Flutter, Firebase, JavaScript

- Led the official software development club** at Korean Minjok Leadership Academy (KMLA), administering and maintaining the student community website using **PHP and JavaScript**.
- Developed and published cross-platform mobile applications for school meal schedules and news distribution using **Flutter and Firebase**, improving campus information access for the entire student body.

KCU (Team Lead) | CPP, SECURITY, WINDOWS API, REACT, ELECTRON

- Spearheaded reverse-engineering and system security research** on the open-source game *AssaultCube*, using **C++ and Windows API** to analyze process memory structures, implementing **trampoline-based OpenGL hooking** for runtime function interception, and developing a **client communicating through shared memory** for inter-process communication.
- Engineered a desktop virtual camera system with dynamic real-time video filters** using **React, Electron, C++, and Windows Media Foundation**, integrating **native Windows APIs, video frame processing, and inter-process communication (IPC)** for low-latency video streaming.
- Managed development of a course management platform integrating the Canvas API**, building services to retrieve and organize **course, assignment, grade, and deadline data** and integrating **calendar APIs** to synchronize academic events into a unified dashboard.